



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science / Sixth Semester – Spring 2024

Subject: Enterprise Application Development

Paper: EC-332

Roll No. 0

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- i. What is Data set in ADO net?
- ii. What are SQL connection methods?
- iii. How to detect a mobile device in jQuery?
- iv. What is the purpose of clone () in data set methods?
- v. What is JSON?
- vi. What is SOAP?

Q.2. Answer the following questions.

(2x15=30)

- a) What is static method of JavaScript? Write a program in JavaScript to calculate the area and circumference of a circle.
- b) What is C# execution handling? Write a C# program to check whether the number is Armstrong or not.



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science / Sixth Semester – Spring 2024

Subject: Artificial Intelligence

Paper: DC-324

Roll No. _____

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

- i. ✓ What is difference between Episodic and Sequential Environment? Explain with the help of an Example.
- ii. ✓ What are included in Problem Definition? Write problem definition of any one toy problem.
- iii. ✓ Which factors determine performance of a problem solving technique?
- iv. ✓ What is a bidirectional Search? How it is useful?
- v. ✓ What is game playing? How it is used as a problem solving technique?
- vi. ✓ What is difference between weak and strong intelligence?

Q.2. Answer the following questions. (3x10=30)

- ✓ 1. What is a Sensor-less Problem? Discuss with the help of an example.
- ✓ 2. Discuss with the help of an example how A* search is better than Greedy Best First Search.
- 3. Give brief answers of following:
 - a) Prove that A* is consistent.
 - b) How can two heuristic functions can be compared?



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Sixth Semester – Spring 2024

Subject: Mobile Application Development

Paper: EC-333

Roll N

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short note on the following:

(6x5=30)

- ✓ 1) What is **activity_main.xml** file?
- 2) What is **AVD** configuration in android app?
- 3) What type of object is return by **getSelectedItem ()**?
- ✓ 4) What is **on Resume ()** method?
- ✓ 5) What is **linear layout**?
- ✓ 6) What is **Array Adapter**?

Q.2. Answer the following questions.

(2x15=30)

1. Write a code to implement **ImageSwitcher** component in **.xml** file.
The **setfactory ()** method of **ImageSwitcher** provide implementation of **Viewfactory** interface implements its unimplemented method returns an **ImageView**.
- ✓ 2. Create a file **CheckBox.xml** and create both default as well as custom check box in **CheckBox.xml** file.



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Sixth Semester – Spring 2024

Subject: Computer Networks

Paper: CC 313

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short note on the following:

(6x5=30)

1. Explain the concept of circuit switching and provide an example scenario where it is commonly used. Contrast it with packet switching?
- ✓ 2. Describe the functions of the physical layer in the OSI model. How does it differ from the data link layer?
- ✓ 3. Discuss the purpose and use of MAC addressing. Provide a practical example of a MAC address?
- ✓ 4. Explain the basics of CIDR (Classless Inter-Domain Routing) and how it impacts IP addressing and routing in the Internet?
- ✓ 5. Define the roles of the transport layer in network communication. Discuss the concepts of connection establishment and flow control?
- ✓ 6. Name two application layer protocols commonly used in modern computer networks and describe their functions?

Q.2. Answer the following questions.

(3x10=30)

1. Discuss the key features and advantages of IPv6 over IPv4. Explain the challenges involved in the transition to IPv6 and potential solutions?
2. Explain the concept of routing and the role of routing protocols in network communication. Compare and contrast two routing protocols used in the Internet?
3. Explore and discuss the latest trends in computer networks, such as SDN (Software-Defined Networking), IoT (Internet of Things), and 5G networks. Explain their significance and potential impact on the industry?



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science / Sixth Semester – Spring 2024
Subject: Parallel & Distributed Computing

Paper: DC-323

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

✓1. Assume the fraction of serial program (f) is 0.34, what is the speedup on (10)

- 2 processors.
- 4 processors.

✓2. Suppose there are four processes including the root, each with arrays as shown in figure below.

array u	Rank
10 11 12 13 14 15 16 17	0
20 21 22 23 24 25 26 27	1
30 31 32 33 34 35 36 37	2
40 41 42 43 44 45 46 47	3

`MPI_Alltoall(u, 2, MPI_INT, v, 2, MPI_INT, MPI_COMM_WORLD);`

After the all to all operation, show how data will be distributed. You need to show the contents of the resultant array at each rank. (10)

✓3. Consider the following matrix A (5 x 5) and vector B (5 x 1) that we intend to multiply. Suppose we have 5 processors and while solving this problem in parallel; we broadcast each element of a vector to respective processor. Also we broadcast each column of a matrix to one respective processor.

Suppose (as given in example), there are some entries of matrix with 0 in them. Assume that we can plan something that can help to avoid such elements (and communication with the processors holding these elements). Which processors will need to communicate to compute the green cell of the resultant matrix? (5)

4	1	2	0	0	3
23	44	6	0	0	1
1	2	5	0	0	1
23	2	55	0	0	1
2	33	4	0	0	2

✓4. Assume you have the following graph. What will be the length of the critical path assuming unit work at each process? (5)



Q.2. Answer the following questions.

1 Suppose you want to map the data in three processes to its given form as below (using All to all v operation). As you know that we need to prepare count and displacement arrays for both the sender and receiver end. As the number of processors is 3, therefore these array will be of size 3. (5)

Process 0			Process 1			Process 2		
Value	Value	Value	Value	Value	Value	Value	Value	Value
2	22	0	33	4	1	24	3	9

↓ MPI_AlltoAllv

Process 0			Process 1			Process 2		
Value	Value	Value	Value	Value	Value	Value	Value	Value
3	11	0	22			0	11	

What will be the contents of count array for processor rank 2 (as a sender)?

- 2 Suppose you want to map the data in three processes to its given form as below (using All to all v operation). As you know that we need to prepare count and displacement arrays for both the sender and receiver end. As the number of processors is 3, therefore these array will be of size 3. (5)

Process 0			Process 1			Process 2		
Value	Value	Value	Value	Value	Value	Value	Value	Value
2	22	0	11	4	3	24	3	9

↓ MPI_AlltoAllv

Process 0			Process 1			Process 2		
Value	Value	Value	Value	Value	Value	Value	Value	Value
3	11	0	22			0	11	

What will be the contents of count array for processor rank 1 (as a receiver)?

- 3 Consider the following code with 3 processors. What will the processor with rank 2 get after scatter operation? (10)

```

1  #include <mpi.h>
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <math.h>
5  int main(int argc, char* argv[])
6  {
7      int rank, size;
8      MPI_Init(&argc, &argv);
9      MPI_Comm_rank(MPI_COMM_WORLD, &rank);
10     MPI_Comm_size(MPI_COMM_WORLD, &size);
11     int buffer[7] = { 10, 4, 11, 52, 6, 70, 73 };
12     int counts[3] = { 3, 2, 5 };
13     int displacements[3] = { 2, 2, 0 };
14     int* RecBuff = (int*)malloc(sizeof(int) * counts[rank]);
15     MPI_Scatterv(buffer, counts, displacements, MPI_INT, RecBuff, counts[rank], MPI_INT, 0, MPI_COMM_WORLD);
16     MPI_Finalize();
17     return 0;
18 }

```

- 4 Derive number of links, arc connectivity, bisection width, and diameter of a the following interconnection networks and fill the table. Your derivation must be supported by logical claims otherwise zero will be granted. Furthermore, the derivation should be generic i.e.; not with respect to some particular number of processors. (10)

Network	Diameter	Bisection width	Arc connectivity	Number of links
Binary tree				
Hypercube				
completely connected graph				